Notes on lecture 17

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1 Vector forms of Green's theorem

Consider $\vec{F} = P\hat{i} + Q\hat{j}$ in 3-dimensional space (that is, $R = 0$). So

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & 0 \end{vmatrix} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \hat{k}.$$

If C is a positively oriented, piecewise smooth, simple closed curve, that bounds a region D, then we have

$$\int_C P dx + Q dy = \oint_C \vec{F} \cdot \vec{dr} = \int_D \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$= \int_D \int \left[\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \hat{k} \right] \cdot \hat{k}\ dA = \int_D \int (\nabla \times \vec{F}) \cdot \hat{k}\ dA \qquad (*)$$

The above equation $(*)$ expresses the line integral of the tangential component of $\vec{F}$ along C as the double integral of the vertical component of $curl \vec{F} (= \nabla \times \vec{F})$ over the region D enclosed by C.
If C is given by the vector equation

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j},\ a \leq t \leq b,$$

then the unit tangent vector is

$$\vec{T}(t) = \frac{x'(t)}{|\vec{r'}(t)|}\hat{i} + \frac{y'(t)}{|\vec{r'}(t)|}\hat{j}.$$

The outward unit normal vector to C is given by

$$\hat{n}(t) = \frac{y'(t)}{|\vec{r'}(t)|}\hat{i} - \frac{x'(t)}{|\vec{r'}(t)|}\hat{j}.$$

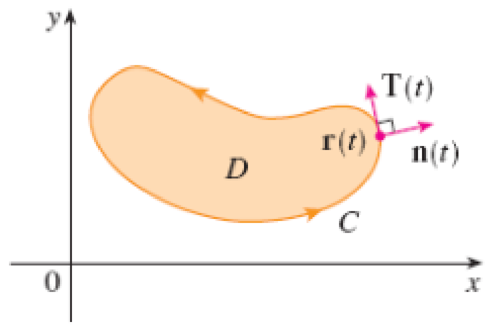

Figure 1.0.1: Vector forms of Green's theorem

Please refer to Figure 1.0.1 for this. Then we have

$$\oint_C \vec{F} \cdot \hat{n} ds = \int_a^b (\vec{F} \cdot \hat{n})(t) |\vec{r'}(t)| dt$$

$$= \int_a^b \left[\frac{P(x(t), y(t))y'(t)}{|\vec{r'}(t)|} - \frac{Q(x(t), y(t))x'(t)}{|\vec{r'}(t)|} \right] |\vec{r'}(t)| dt$$

$$= \int_a^b P(x(t), y(t))y'(t)dt - Q(x(t), y(t))x'(t)dt = \int_C Pdy - Qdx$$

$$= \int_D \int \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA \text{ (by Green's theorem).}$$

But the integrand in this double integral is just the divergence of $\vec{F}$. So we have a **vector form of Green's theorem**:

$$\oint_C \vec{F} \cdot \hat{n} ds = \int_D \int div \vec{F}(x, y) dA.$$

This version says that the line integral of the normal component of $\vec{F}$ along C is equal to the double integral of the divergence of $\vec{F}$ over the region D enclosed by C.

2 Parametric surfaces and their areas

2.1 Parametric curves

Recall that in the case of parametric curves, we had considered an interval $[a, b]$, and t was a variable such that $a \leq t \leq b$. Then we defined some functions $x(t), y(t), z(t)$ from $[a, b]$ to the real line $\mathbb{R}$. As the variable t runs through $[a, b]$, for each value of t, we are going to get a point $(x(t), y(t), z(t))$ in space. If we plot all these points in space (as t varies in the interval $[a, b]$), we are going to get a curve. Corresponding to an arbitrary point $P = (x(t), y(t), z(t))$ lying on the curve so generated, we can always define a vector $\vec{r}(t)$ whose components are $x(t), y(t)$, and $z(t)$.

2.2 Parametric surfaces

In case of parametric curves, we had considered an interval $[a, b]$. Now, let us consider a domain D ($\subseteq \mathbb{R}^2$) in a 2 dimensional plane (which we call the uv-plane). Define 3 functions $x(u, v), y(u, v)$, and $z(u, v)$ from D to $\mathbb{R}$. Then corresponding to each point $(u, v) \in D$, we are going to obtain a point in the 3 dimensional space. Now if we consider the union of all such points in the 3 dimensional space, then we are going to obtain a surface S. This surface is going to be called a **parametric surface**, because it is generated by the 2 parameters u and v. The equations

$$x = x(u, v), \ y = y(u, v), \ z = z(u, v)$$

are called the **parametric equations** for the surface S. So basically, corresponding to each point $(u, v) \in D$, we are going to get a point $P = (x(u, v), y(u, v), z(u, v))$ on the surface S. Now if we joing the origin with the point P, we are going to get a vector, which we denote by $\vec{r}(u, v)$. Here,

$$\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k}.$$

Figure 2.2.1 explains this phenomenon.

Remark 2.2.1. Earlier in the case of parametric curves, the vector $\vec{r}(t)$ was defined in terms of a single parameter t, but in this case (of parametric surface) the vector $\vec{r}(u, v)$ is defined in terms of 2 variables u and v.

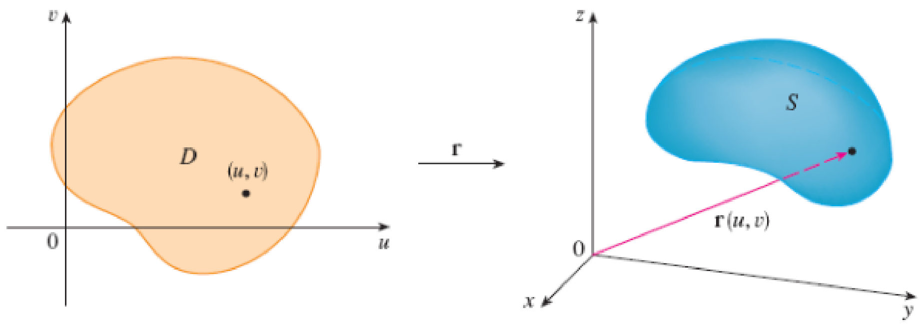

Figure 2.2.1: A parametric surface being generated

Example 2.2.2. Let $\vec{r}(u, v) = u\hat{i} + u\cos v\hat{j} + u\sin v\hat{k}$. Find the surface given by the above parametric representation.
Clearly,

$$\vec{r}(u, v) = x\hat{i} + y\hat{j} + z\hat{k},$$

where $x = u, y = u\cos v$, and $z = u\sin v$.
Observe that $y^2 + z^2 = u^2 = x^2$. So the equation of the surface is given by $x^2 = y^2 + z^2$. $\square$

Example 2.2.3. Find a parametric representation for the cylinder

$$x^2 + y^2 = 4, \;\; 0 \leq z \leq 1.$$

A parametric representation is given by

$$\vec{r}(\theta, z) = x\hat{i} + y\hat{j} + z\hat{k},$$

where

$$x = 2\cos\theta, y = 2\sin\theta, z = z, 0 \leq \theta \leq 2\pi, 0 \leq z \leq 1.$$

$\square$

Example 2.2.4. Find a vector function that represents the elliptic paraboloid $z = x^2 + 2y^2$.
The required vector function is given by

$$\vec{r}(x, y) = x\hat{i} + y\hat{j} + (x^2 + 2y^2)\hat{k}.$$

$\square$

Remark 2.2.5. Whenever we have a surface which is the graph of a function of 2 variables, say given by $z = f(x, y)$, then the job of finding out a vector function that represents the surface is easy (as can be seen in the above example). It is just given by

$$\vec{r}(x, y) = x\hat{i} + y\hat{j} + f(x, y)\hat{k}.$$

2.3 The grid curves

Now we are going to see how constant u, v lines (that is, lines of the form $u = u_0$ and $v = v_0$) in the region D of the uv-plane will correspond to some curves over the surface S called "grid curves".
Consider a domain D in the uv-plane. Let $(u_0, v_0) \in D$. Consider the lines $u = u_0$ and $v = v_0$ in D. Let S denote the parametric surface generated by all the points in D via

$$\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k}.$$

Let us now see how the images of the lines $u = u_0$ and $v = v_0$ in D will look like in the surface S. Observe that

$$\vec{r}(u_0, v) = x(u_0, v)\hat{i} + y(u_0, v)\hat{j} + z(u_0, v)\hat{k},$$

which is essentially a function of the single parameter v. A single parameter function will generate a curve in the 3 dimensional space. So, the line $u = u_0$ in D will generate a curve (say C_1) on the surface S. Similarly, corresponding to the line $v = v_0$ in D, we will get a curve (say, C_2) on the surface S. Such curves are called **grid curves**. The point of intersection of the curves C_1 and C_2 is going to have position vector $\vec{r}(u_0, v_0)$. Please see Figure 2.3.1 for an illustration.

If we generate a grid on D by drawing constant u and v lines, then it will give us a grid (made up of curves) on the surface S. Such curves are called grid curves (as already mentioned earlier). Figure 2.3.2 shows some grid curves on a surface.

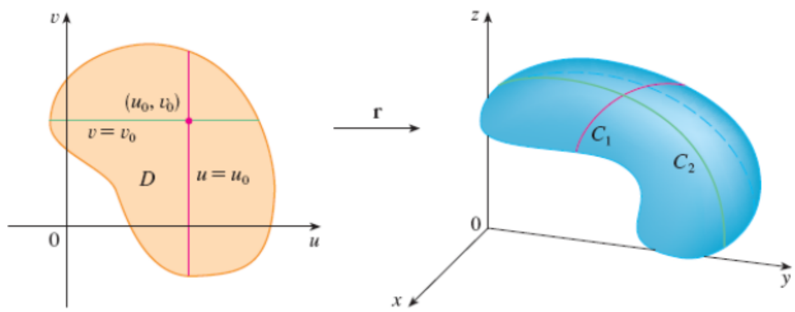

Figure 2.3.1: Grid curves

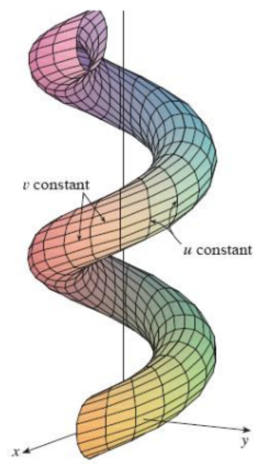

Figure 2.3.2: Grid curves (an example)

Example 2.3.1. Find a parametric representation of the sphere $x^2+y^2+z^2 = a^2$.
Using spherical coordinate system, we can easily see that a parametric representation of the above sphere is given by

$$\vec{r}(\theta, \phi) = x\hat{i} + y\hat{j} + z\hat{k},$$

where

$$x = a\sin\phi\cos\theta, y = a\sin\phi\sin\theta, z = a\cos\phi$$

and

$$0 \le \theta \le 2\pi, 0 \le \phi \le \pi.$$

Figure 2.3.3 shows some grid curves on this sphere. $\square$

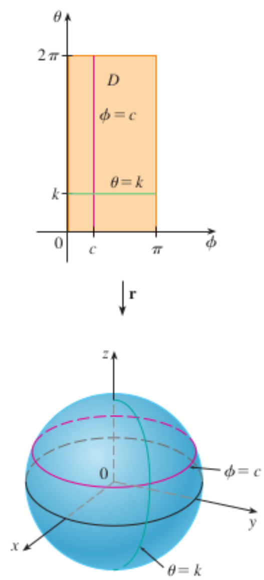

Figure 2.3.3: Figure for Example 2.3.1

2.4 Equation of tangent planes to parametric surfaces

We now find out the tangent plane to a parametric surface S traced out by a vector function

$$\vec{r}(u,v) = x(u,v)\hat{i} + y(u,v)\hat{j} + z(u,v)\hat{k}$$

at a point P_0 with position vector $\vec{r}(u_0, v_0)$. If we keep u constant by putting $u = u_0$, then $\vec{r}(u_0, v)$ becomes a vector function of the single parameter v and defines a grid curve C_1 lying on S, see Figure 2.4.1.

The tangent vector to C_1 at P_0 is obtained by taking the partial derivative of $\vec{r}$ with respect to v:

$$\vec{r_v} = \frac{\partial x}{\partial v}(u_0, v_0)\hat{i} + \frac{\partial y}{\partial v}(u_0, v_0)\hat{j} + \frac{\partial z}{\partial v}(u_0, v_0)\hat{k}.$$

Similarly, if we keep v constant by putting $v = v_0$, we get a grid curve C_2 given by $\vec{r}(u, v_0)$ that lies on S, and its tangent vector at P_0 is

$$\vec{r_u} = \frac{\partial x}{\partial u}(u_0, v_0)\hat{i} + \frac{\partial y}{\partial u}(u_0, v_0)\hat{j} + \frac{\partial z}{\partial u}(u_0, v_0)\hat{k}.$$

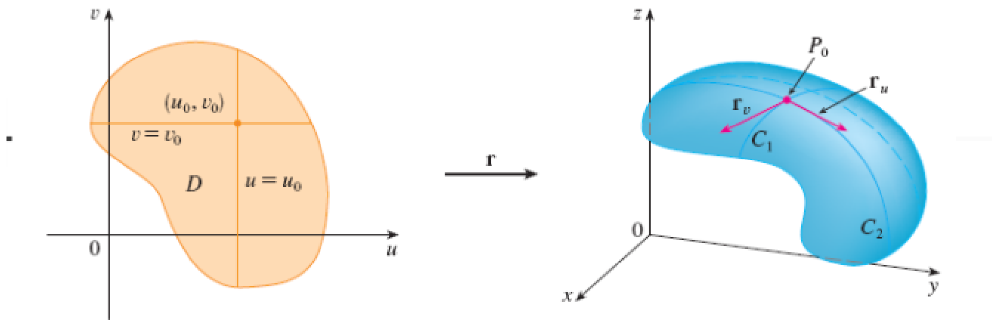

Figure 2.4.1: Tangent planes to parametric surfaces

Now if we consider the cross product of the two tangent vectors $\vec{r_u}(u_0, v_0)$ and $\vec{r_v}(u_0, v_0)$, it will give us the normal to the tangent plane to the surface S at the point P_0. If we know the point $P_0 = (x_0, y_0, z_0)$ and the components $< a, b, c >$ of the normal, then the equation of the tangent plane will be

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

Definition 2.4.1. If $\vec{r_u} \times \vec{r_v} \neq \vec{0}$, then the surface S is called **smooth** (it has no "corners"). For a smooth surface, the tangent plane is the plane that contains the tangent vectors $\vec{r_u}$ and $\vec{r_v}$, and the vector $\vec{r_u} \times \vec{r_v}$ is a normal vector to the tangent plane. □

Example 2.4.2. Find the tangent plane to the surface with parametric equation

$$x = u^2, y = v^2, z = u + 2v$$

at the point $(1, 1, 3)$.
Observe that since $x = y = 1$ and $z = 3$, therefore we must have $u = v = 1$. So $u_0 = v_0 = 1$ here. We have

$$\vec{r}(u, v) = u^2\hat{i} + v^2\hat{j} + (u + 2v)\hat{k}.$$

This implies that

$$\vec{r_u} = 2u\hat{i} + \hat{k}$$

and

$$\vec{r_v} = 2v\hat{j} + 2\hat{k}.$$

Therefore,

$$\vec{r_u}(1,1) = 2\hat{i} + \hat{k} \text{ and } \vec{r_v}(1,1) = 2\hat{j} + 2\hat{k}.$$

So the normal is $\vec{r_u} \times \vec{r_v}$, where

$$\vec{r_u} \times \vec{r_v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 1 \\ 0 & 2 & 2 \end{vmatrix} = -2\hat{i} - 4\hat{j} + 4\hat{k}.$$

Hence the equation of the tangent plane at $(1,1,3)$ to the surface is

$$-2(x-1) - 4(y-1) + 4(z-3) = 0.$$

Figure 2.4.2 shows the surface along with the tangent plane at $(1,1,3)$. $\square$

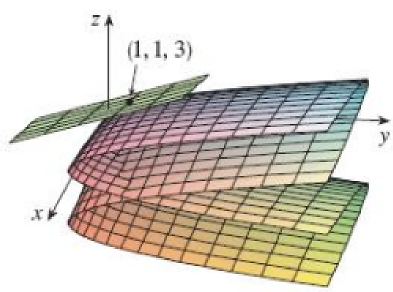

Figure 2.4.2: Figure for Example 2.4.2

2.5 Surfaces of revolution

Consider a function $y = f(x)$ defined on the interval $[a, b]$ along the x-axis. The graph of this function is a curve in the xy-plane. If we rotate this curve about the x-axis, then it will generate a surface. If we now take the cross section of this surface by cutting it with a plane parallel to the yz-plane, then it is going to generate a circle of radius $f(x)$. Let $P(x, y, z)$ be a point on the surface. Let us now join the point $(x, 0, 0)$ with P. Then the distance between P and the point $(x, 0, 0)$ will be $f(x)$. Now if the vector joining P with the point $(x, 0, 0)$ makes an angle θ with the line on the xy-plane parallel to the y-axis, then the y-coordinate of P will be given by $y = f(x)\cos\theta$ and the z-coordinate of P will be given by $z = f(x)\sin\theta$. Please see Figure 2.5.1 for an illustration.

In this way, we have found out a parametrization of the surface of revolution

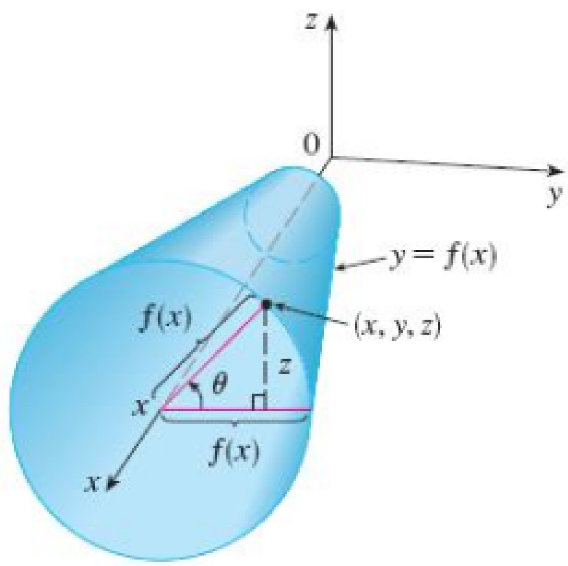

Figure 2.5.1: Surfaces of revolution

in terms of x and θ. This parametrization is given by

$$\vec{r}(x, \theta) = x\hat{i} + f(x)\cos\theta\hat{j} + f(x)\sin\theta\hat{k}.$$

We are going to use this concept later on to find the area of a surface of revolution.

Exercise: Find parametric equations for the surface generated by rotating the curve $y = \sin x$, $0 \leq x \leq 2\pi$, about the x-axis. Figure 2.5.2 depicts the surface.

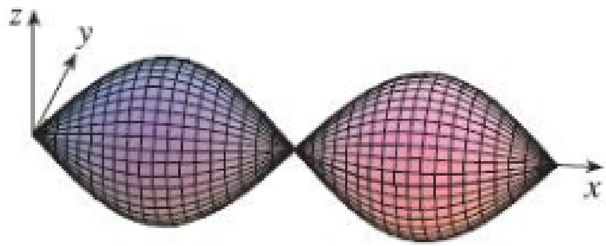

Figure 2.5.2: Surface generated by rotating the curve $y = \sin x$ along the x-axis

2.6 Area of a parametric surface

Back to change of variables

In change of variables (studied earlier), we had converted a rectangular domain S in the uv-plane into a curved rectangular domain R in the xy-plane. Please refer to Figure 2.6.1 for this.

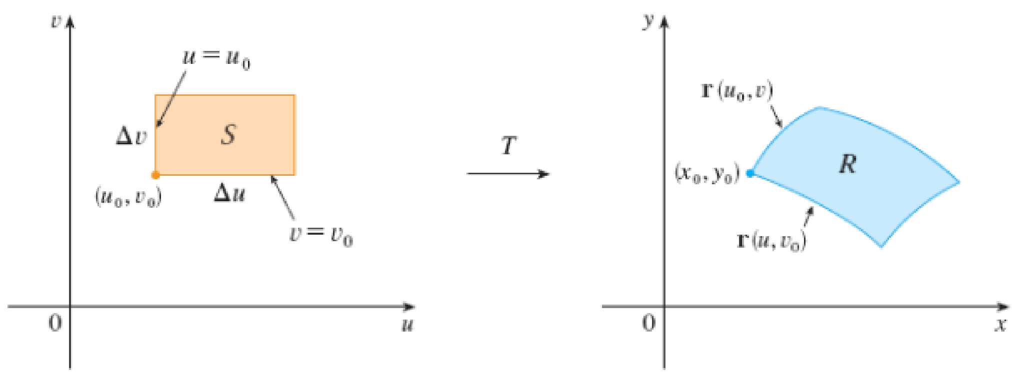

Figure 2.6.1: Change of variables

If we consider the tangent lines to the bottom and the left curves of the region R (refer to the right hand side picture of Figure 2.6.1) at the point (x_0, y_0), then those 2 tangent lines will generate a parallelogram, as shown in Figures 2.6.2 and 2.6.3. Then we approximated the area of the blue surface

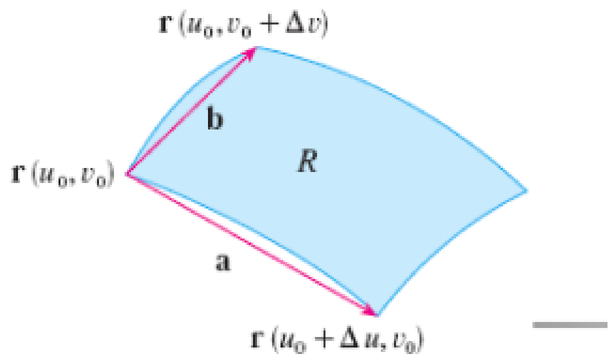

Figure 2.6.2: The 2 tangent lines

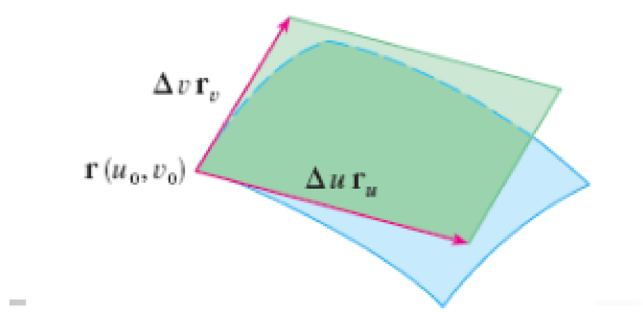

Figure 2.6.3: The parallelogram

(in Figure 2.6.3) by the help of the area of this (green colored) parallelogram, which was $|\Delta u\vec{r}_u \times \Delta v\vec{r}_v|$. We are going to use the same concept for finding out the area of a parametric surface.

<u>Surface area of a general parametric surface</u>

We now define the surface area of a general parametric surface given by the equation

$$\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k}.$$

For simplicity, we start by considering a surface whose parameter domain D is a rectangle, and we divide it into subrectangles R_{ij}. Let us choose (u_i^*, v_j^*) to be the lower left corner of R_{ij}. The part S_{ij} of the surface S that corresponds to R_{ij} is called a **patch**. This patch has the point P_{ij} with position vector $\vec{r}(u_i^*, v_j^*)$ as one of its corners. See Figure 2.6.4 for an illustration.

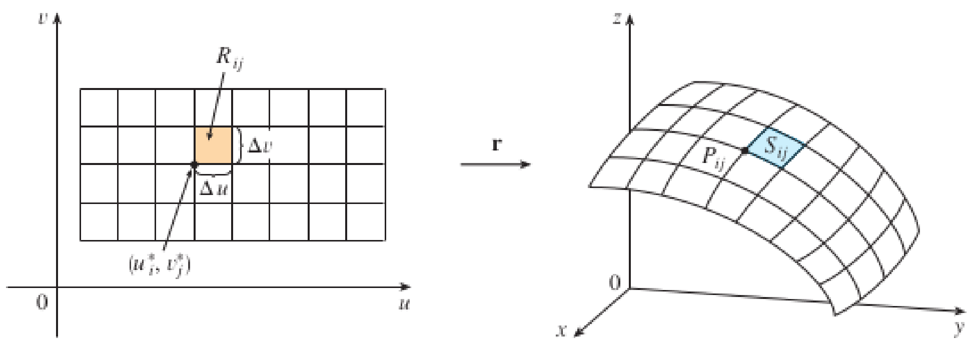

Figure 2.6.4: The patch S_{ij} corresponding to R_{ij}

Let $\vec{r}_u^* = \vec{r}_u(u_i^*, v_j^*)$ and $\vec{r}_v^* = \vec{r}_v(u_i^*, v_j^*)$ be the tangent vectors at P_{ij}.

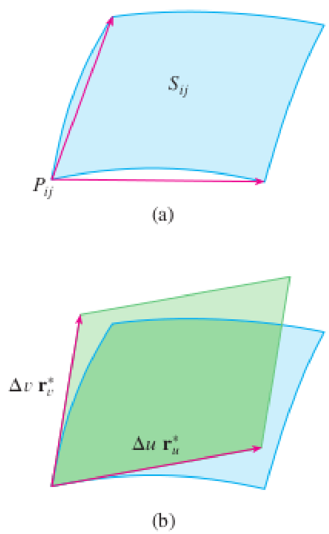

Figure 2.6.5: Approximating the area of S_{ij}

The picture in part (a) of Figure 2.6.5 shows how the two edges of the patch that meet at P_{ij} can be aprroximated by vectors. These vectors in turn, can be approximated by the vectors $\Delta u \vec{r}_u^*$ and $\Delta v \vec{r}_v^*$ because partial derivatives can be approximated by difference quotients. So we approximate the area of S_{ij} by the area of the parallelogram determined by the vectors $\Delta u \vec{r}_u^*$ and $\Delta v \vec{r}_v^*$. This parallelogram is shown in part (b) of Figure 2.6.5 and it lies in the tangent plane to S at P_{ij}. The area of this parallelogram is

$$|(\Delta u \vec{r}_u^*) \times (\Delta v \vec{r}_v^*)| = |\vec{r}_u^* \times \vec{r}_v^*| \Delta u \Delta v$$

and so an approximation to the area of S is

$$\sum_{i=1}^{m} \sum_{j=1}^{n} |\vec{r}_u^* \times \vec{r}_v^*| \Delta u \Delta v.$$

As we increase the number of subrectangles, this approximation gets better and better and we recognize the above double sum as a Riemann sum for the double integral $\int_D \int |\vec{r}_u \times \vec{r}_v| dudv$. This motivates the following definition.

Definition 2.6.1. If a smooth parametric surface S is given by the equation

$$\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k},$$

where $(u, v) \in D$, and S is covered just once as (u, v) ranges through the parameter domain D, then the **surface area** of S is given by

$$A(S) = \int_D \int |\vec{r}_u \times \vec{r}_v| dA,$$

where

$$\vec{r}_u = \frac{\partial x}{\partial u}\hat{i} + \frac{\partial y}{\partial u}\hat{j} + \frac{\partial z}{\partial u}\hat{k}$$

and

$$\vec{r}_v = \frac{\partial x}{\partial v}\hat{i} + \frac{\partial y}{\partial v}\hat{j} + \frac{\partial z}{\partial v}\hat{k}.$$

□

Example 2.6.2. Find the surface area of a sphere of radius a.
Using spherical coordinate system, we can easily see that a parametric representation of the above sphere is given by

$$\vec{r}(\theta, \phi) = x\hat{i} + y\hat{j} + z\hat{k},$$

where

$$x = a\sin\phi\cos\theta, y = a\sin\phi\sin\theta, z = a\cos\phi$$

and

$$0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \pi.$$

Therefore,

$$\vec{r}_\theta =< -a\sin\phi\sin\theta, a\sin\phi\cos\theta, 0 >$$

and

$$\vec{r}_\phi =< a\cos\phi\cos\theta, a\cos\phi\sin\theta, -a\sin\phi > .$$

It is your exercise to check that

$$\vec{r}_\theta \times \vec{r}_\phi =< -a^2\sin^2\phi\cos\theta, -a^2\sin^2\phi\sin\theta, -a^2\sin\phi\cos\phi > .$$

Therefore,

$$|\vec{r}_\theta \times \vec{r}_\phi| = a^2\sin\phi.$$

Hence the required area

$$= \int_D \int |\vec{r}_\theta \times \vec{r}_\phi| dA = \int_D \int a^2 \sin\phi d\theta d\phi,$$

where $D = \{(\theta, \phi) | 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \pi\}$. It can be easily calculated that this integral equals $4\pi a^2$. □

2.7 An observation

Recall that when we considered surfaces of revolution, we make the curve $y = f(x)$ lying on the xy-plane rotate about the x-axis. Then it generated a surface which is given by

$$\vec{r}(x,\theta) =< x, f(x)\cos\theta, f(x)\sin\theta >,$$

where θ is as explained in §1.5 above. So

$$\vec{r}_x =< 1, f'(x)\cos\theta, f'(x)\sin\theta >$$

and

$$\vec{r}_\theta =< 0, -f(x)\sin\theta, f(x)\cos\theta > .$$

It can be easily checked that

$$\vec{r}_x \times \vec{r}_\theta = f(x)f'(x)\hat{i} - f(x)\cos\theta\hat{j} - f(x)\sin\theta\hat{k}.$$

Therefore,

$$|\vec{r}_x \times \vec{r}_\theta| = f(x)\sqrt{1 + [f'(x)]^2}.$$

Hence the area of the surface of revolution

$$= \int_D\int f(x)\sqrt{1 + [f'(x)]^2}dA,$$

where $dA = dxd\theta$ and D is a rectangular region in the (x,θ)-plane. Observe that this is exactly what we had obtained in earlier lectures while studying areas of surfaces of revolution.

2.8 Surface area of the graph of a function

Sometimes, a surface can be the graph of a function of 2 variables, say $z = f(x,y)$. In that case, we are simply going to use the parametrization

$$\vec{r}(x,y) = x\hat{i} + y\hat{j} + f(x,y)\hat{k}.$$

Then

$$\vec{r}_x = \hat{i} + f_x\hat{k}$$

and

$$\vec{r}_y = \hat{j} + f_y\hat{k}.$$

Therefore,

$$\vec{r}_x \times \vec{r}_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{vmatrix} = -f_x\hat{i} - f_y\hat{j} + \hat{k}$$

and

$$|\vec{r}_x \times \vec{r}_y| = \sqrt{1 + f_x^2 + f_y^2}.$$

Hence we have from $(*)$ that

$$A(S) = \int_D \int \sqrt{1 + z_x^2 + z_y^2} dA,$$

where $z = f(x, y)$.

Remark 2.8.1. The surface $z = f(x, y)$ can be considered as a level surface given by $F(x, y, z) = 0$, where $F(x, y, z) = z - f(x, y)$. Then ∇F is the normal to the surface $F(x, y, z) = 0$. So

$$\nabla F =< F_x, F_y, F_z >=< -f_x, -f_y, 1 > .$$

Remark 2.8.2. Recall that for deriving formula $(*)$, we considered a tangent plane at a point, and at that point, the normal vector came out to be $\vec{r}_u \times \vec{r}_v$. Therefore, the unit normal to the parametric surface at that point is given by

$$\hat{n} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}.$$

Example 2.8.3. Find the area of the part of the paraboloid $z = x^2 + y^2$ that lies under the plane $z = 9$. Figure 2.8.1 shows the surface (in blue color).

The surface has been given to us as a function of x and y. So a parametrization of the surface is:

$$\vec{r}(x, y) = x\hat{i} + y\hat{j} + (x^2 + y^2)\hat{k}.$$

Here the parameters are x and y. For the given surface, the range of these parameters is given by $(x, y)|x^2 + y^2 \leq 9$. Therefore

$$A(S) = \int_D \int \sqrt{1 + z_x^2 + z_y^2} dA,$$

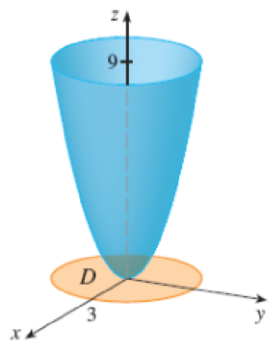

Figure 2.8.1: Figure for Example 2.8.3

where $z = x^2 + y^2$ and $D = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 \leq 9\}$. So we get

$$A(S) = \int_D \int \sqrt{1 + 4(x^2 + y^2)} dA.$$

The parameter domain D is not a rectangle here. It is a disk. So it is better to use polar coordinate system for the above integration. Hence

$$A(S) = \int_{r=0}^{3} \int_{\theta=0}^{2\pi} \sqrt{1 + 4r^2} r dr d\theta.$$

The rest of the calculation is left to the reader as an exercise. $\square$

3 Surface integrals

We have already done multiple integrals earlier. In multiple integrals, we had some planar region/domain D in the xy-plane. [Initially, we had considered a rectangular domain, but later on we extended the concept to any general region.] We subdivided D by some rectangular grid and then (assuming that the graph of the integrand function f lies above the xy-plane) we had some columns above each subrectangle in D of height $f(x_i^*, y_j^*)$, where (x_i^*, y_j^*) was a point in the subrectangle in D. If the dimensions of the subrectangle in D were Δx_i and Δy_j, then we took limit (as $m, n \to \infty$) of the double sum

$$\sum_{i=1}^{n} \sum_{j=1}^{m} f(x_i^*, y_j^*) \Delta x_i \Delta y_j$$

and declared that limit to be the double integral of f over D.

For evaluating surface integrals, we are going to proceed (in a different way) as follows:

Suppose that a surface S has a vector equation

$$\vec{r}(u,v) = x(u,v)\hat{i} + y(u,v)\hat{j} + z(u,v)\hat{k},\ (u,v) \in D.$$

We first assume that the parameter domain D is a rectangle and we divide it into subrectangles R_{ij} having dimensions Δu and Δv. Then the surface S is divided into corresponding patches S_{ij} as shown in Figure 3.0.1.

We evaluate f at a point P_{ij}^* in each patch, multiply by the area ΔS_{ij} of

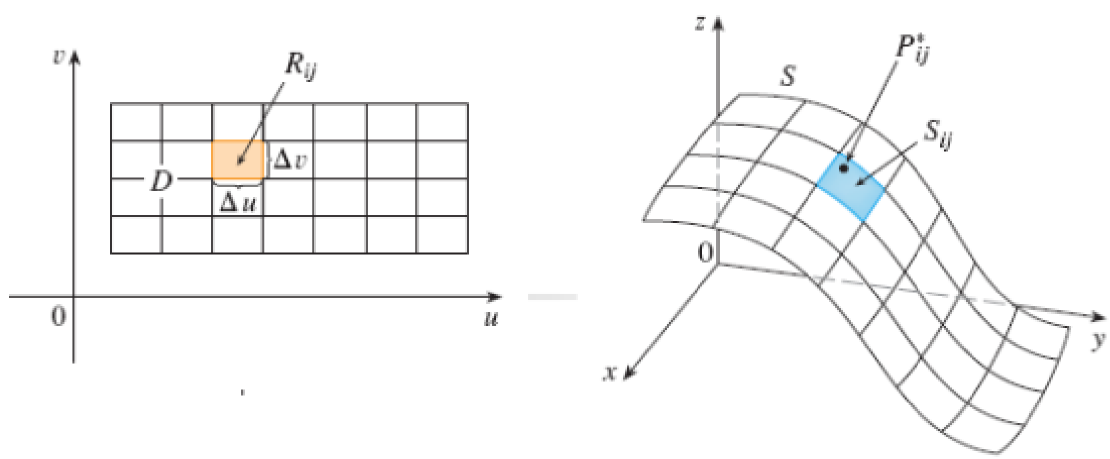

Figure 3.0.1: Surface integrals

the patch, and form the Riemann sum

$$\sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij}^*)\Delta S_{ij}.$$

Then we take the limit as $m, n \to \infty$ and define the **surface integral of** f **over the surface** S as

$$\int_S\int f(x,y,z)dS = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij}^*)\Delta S_{ij}. \qquad (**)$$

To evaluate the surface integral in equation ($**$), we approximate the patch area ΔS_{ij} by the area of an approximating parallelogram in the tangent plane. In our earlier lectures (on surface area), we made the approximation

$$\Delta S_{ij} \approx |\vec{r}_u \times \vec{r}_v| \Delta u \Delta v,$$

where

$$\vec{r}_u = \frac{\partial x}{\partial u}\hat{i} + \frac{\partial y}{\partial u}\hat{j} + \frac{\partial z}{\partial u}\hat{k}$$

and

$$\vec{r}_v = \frac{\partial x}{\partial v}\hat{i} + \frac{\partial y}{\partial v}\hat{j} + \frac{\partial z}{\partial v}\hat{k}$$

are the tangent vectors at a corner of S_{ij}. Therefore, we have

$$\int_S\int f(x,y,z)dS = \int_D\int f(\vec{r}(u,v))|\vec{r}_u \times \vec{r}_v| dA. \qquad (+)$$

Remark 3.0.1. It can be shown that formula $(+)$ holds even when D is not a rectangle, provided the components of $\vec{r}_u$ and $\vec{r}_v$ are continuous, $\vec{r}_u$ and $\vec{r}_v$ are nonzero and nonparallel in the interior of D.

Remark 3.0.2. In equation $(+)$ above, f is a function of 3 variables x, y, z, whereas in case of double integrals, f was a function of 2 variables only.

Exercise: Compute the surface integral $\int_S \int x^2 dS$, where S is the unit sphere $x^2 + y^2 + z^2 = 1$.

3.1 Surface integral of the graph of a function

If the surface S is given by $z = g(x,y)$, then we have

$$\int_S\int f(x,y,z)dS = \int_D\int f(x,y,g(x,y))\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}dA.$$

Note that in the above formula, the expression $\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}$ is nothing but $|\vec{r}_x \times \vec{r}_y|$.

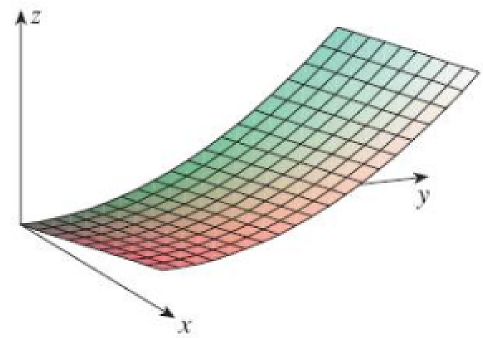

Figure 3.1.1: Figure for Example 3.1.1

Example 3.1.1. Evaluate $\int_S \int y dS$, where S is the surface $z = x + y^2$, $0 \leq x \leq 1$ and $0 \leq y \leq 2$. See Figure 3.1.1 above for a picture of the surface S.

Here the surface S is the graph of a function of 2 variables x and y. Also from the ranges of x and y, we can say that the region D of integration is a rectangle here. So we have

$$\int_S \int y dS = \int_D \int y\sqrt{1 + z_x^2 + z_y^2} dA$$

$$= \int_D \int y\sqrt{1 + 1 + 4y^2} dA = \frac{1}{8} \int_{x=0}^{1} \int_{y=0}^{2} 8y\sqrt{2 + 4y^2} dy dx.$$

The last integral above is in variable separable form. So it is easy to compute the integral now. The rest of the calculation is left to the reader as an exercise. □

3.2 Mass and center of mass

if a thin sheet (say, of aluminum foil) has the shape of a surface S and the density (mass per unit area) at the point (x, y, z) is $\rho(x, y, z)$, then the total **mass** of the sheet is

$$m = \iint_S \rho(x, y, z)\, dS$$

and the **center of mass** is $(\bar{x}, \bar{y}, \bar{z})$, where

$$\bar{x} = \frac{1}{m}\iint_S x\rho(x, y, z)\, dS \qquad \bar{y} = \frac{1}{m}\iint_S y\rho(x, y, z)\, dS \qquad \bar{z} = \frac{1}{m}\iint_S z\rho(x, y, z)\, dS$$

3.3 Surface integral for piecewise smooth surfaces

If S is a piecewise smooth surface, that is, a finite union of smooth surfaces $S_1, S_2, \ldots, S_n$ that intersect only along their boundaries, then the surface integral of f over S is defined by

$$\int_S\int f(x,y,z)dS = \sum_{i=1}^{n}\int_{S_i}\int f(x,y,z)dS.$$

Example 3.3.1. Evaluate $\int_S\int zdS$, where S is the surface whose sides S_1 are given by the cylinder $x^2+y^2=1$, whose bottom S_2 is the disk $x^2+y^2 \leq 1$ in the plane $z=0$, and whose top S_3 is the part of the plane $z=1+x$ that lies above S_2. Figure 3.3.1 shows the surface S.

Clearly

$$\int_S\int zdS = \sum_{i=1}^{3}\int_{S_i}\int zdS.$$

On S_2, we have $z=0$. So $\int_{S_2}\int zdS = 0$.
On S_3, we have $z=1+x$. Therefore

$$\vec{r}(x,y) = x\hat{i} + y\hat{j} + (1+x)\hat{k}$$

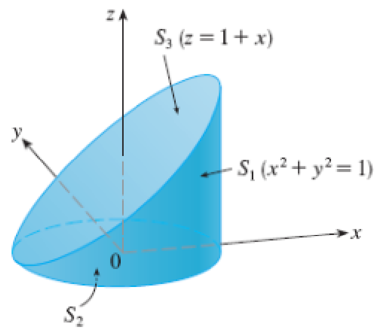

Figure 3.3.1: Figure for Example 3.3.1

and

$$\vec{r}_x \times \vec{r}_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} = -\hat{i} + \hat{k}.$$

This implies that $|\vec{r}_x \times \vec{r}_y| = \sqrt{2}$. Therefore

$$\int_{S_3}\int zdS = \int_{D_3}\int \sqrt{2}zdA = \int_{D_3}\int \sqrt{2}(1+x)dA,$$

where $D_3 = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 \leq 1\}$. So

$$\int_{S_3}\int zdS = \int_{r=0}^{1}\int_{\theta=0}^{2\pi} \sqrt{2}(1 + r\cos\theta)rdrd\theta$$

The rest of the calculation for S_3 is left to the reader as an exercise.
On S_1, we have

$$\vec{r}(\theta, z) = \cos\theta\hat{i} + \sin\theta\hat{j} + z\hat{k}.$$

Here we are taking the parameters to be θ and z because the surface is a cylinder, and so we can take the cylindrical polar coordinates. Therefore

$$\int_{S_1}\int zdS = \int_{D_1}\int z|\vec{r}_\theta \times \vec{r}_z|dA,$$

where $D_1 = \{(\theta, z) | 0 \leq \theta \leq 2\pi, 0 \leq z \leq 1 + \cos\theta\}$. Here the upper limit of the parameter z is $1 + \cos\theta$ because on S_1, the range of the parameter z is

from 0 to $1 + x$, and $x = \cos\theta$ under the transformation used for cylindrical polar coordinates.
The rest of the calculation is left to the reader as an exercise. $\square$

——————-END———————